Foundational Model for NMR Metabolomics

Group Meeting 27th Feb 2026

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Goal

- Build a Foundational model for NMR Metabolomics data
- Why?
  - To classify normal vs abnormal or similar data
  - Maybe help in finding Biomarkers
- What are solving?
  - Deep Neural nets (DNNs) require lots of data to train.
  - Metabolomic datasets are usually very small 10s or 100s of samples

Masked Spectra Modelling (MSM)

- **Task:** hide 15–50% of ppm bins (scattered), **reconstruct** them.

Why it helps later?

- Forces model to learn **multiplet shapes**, **J-patterns**, and **baseline behavior**.
- Yields **features that generalize** to tiny labeled sets (disease/control).
- Acts as a **regularizer** during fine-tuning (less overfitting).

Loss (simple & strong):

- MSE on Masked (0.7) + Unmasked (0.3)

Data Download

- Metabolights
- Contains CPMG and NOESY spectra of many experiments from across the world
- Separated them into – Serum and Plasma
- Total 2146 usable CPMG Serum spectra

Preprocessing

Process all the spectra on TopSpin

- Serial processing - lb 1; si 2*td; em; ft; apk
- This usually makes all the spectra 128k points long

Read all the bruker ‘fid’ files using nmrGlue

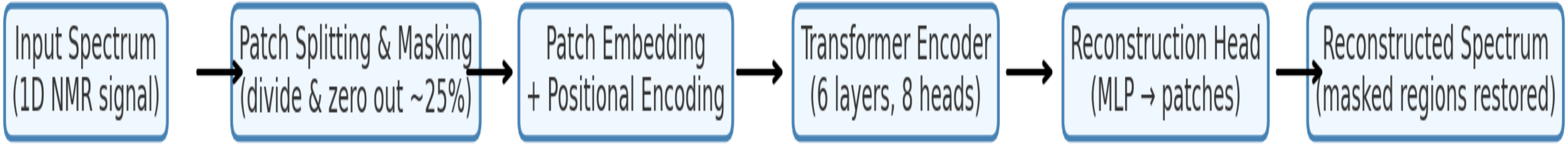
Find the rightmost peak and align spectra using that as reference

Water suppression: points 62500 – 68000 made equal to zero

Remove duplicated spectra

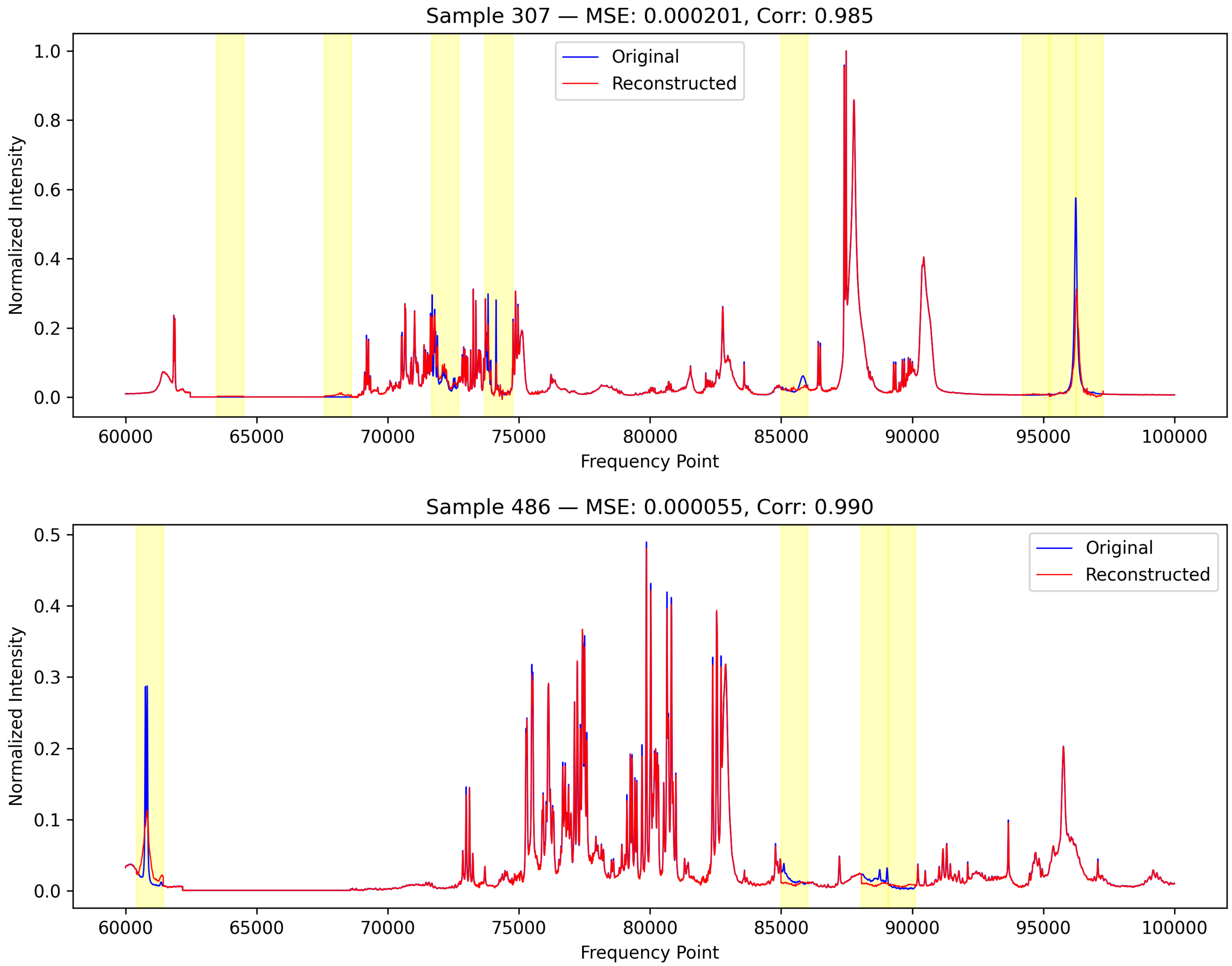
Normalise intensities to between 0-1

Model Architecture



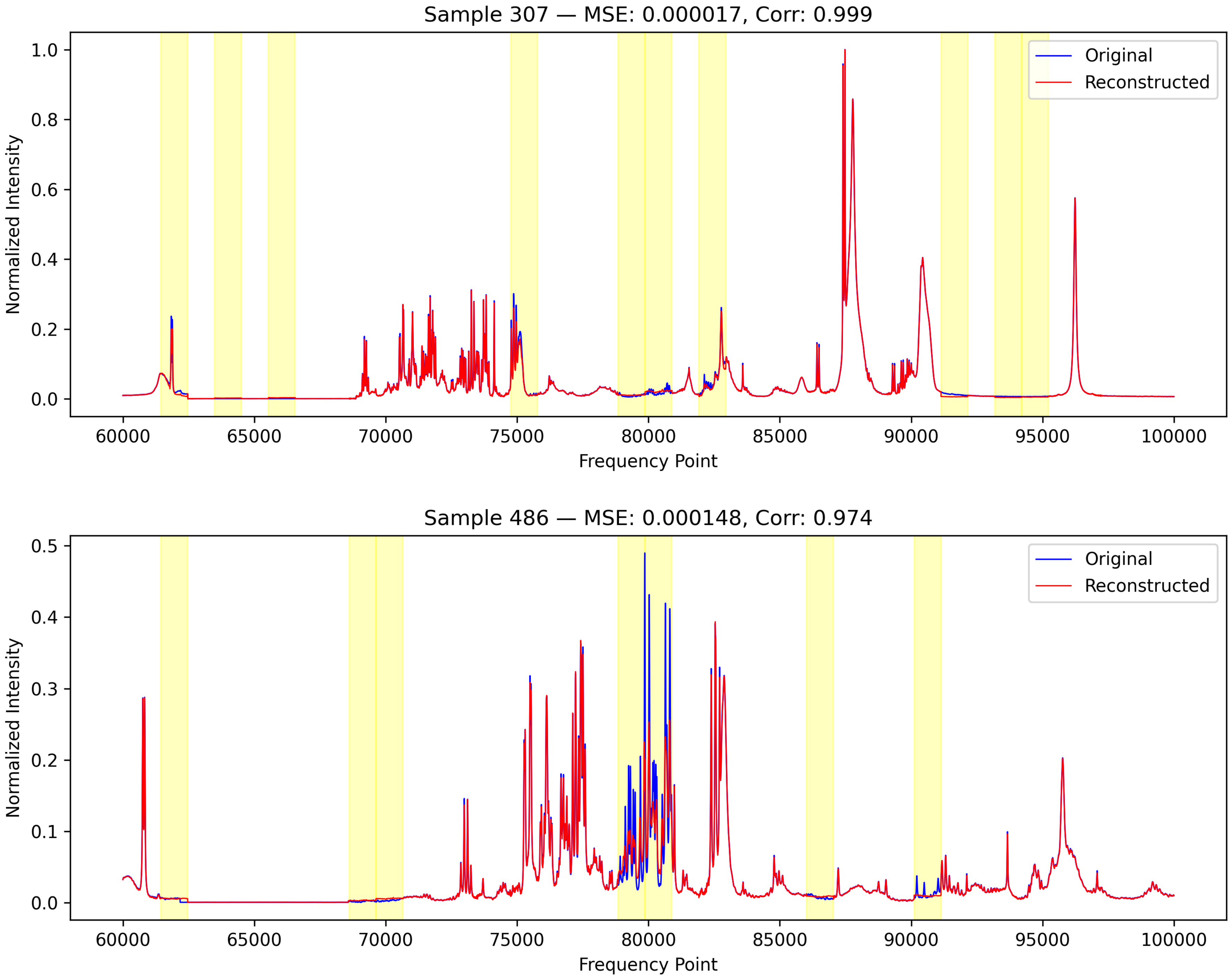
Performance on Train Data

Metric	25% Mask	35% Mask	50% Mask
Overall MSE	$3.28 \times 10-5$	$3.18 \times 10-5$	$5.04 \times 10-5$
Overall R2	0.9805	0.9813	0.9695



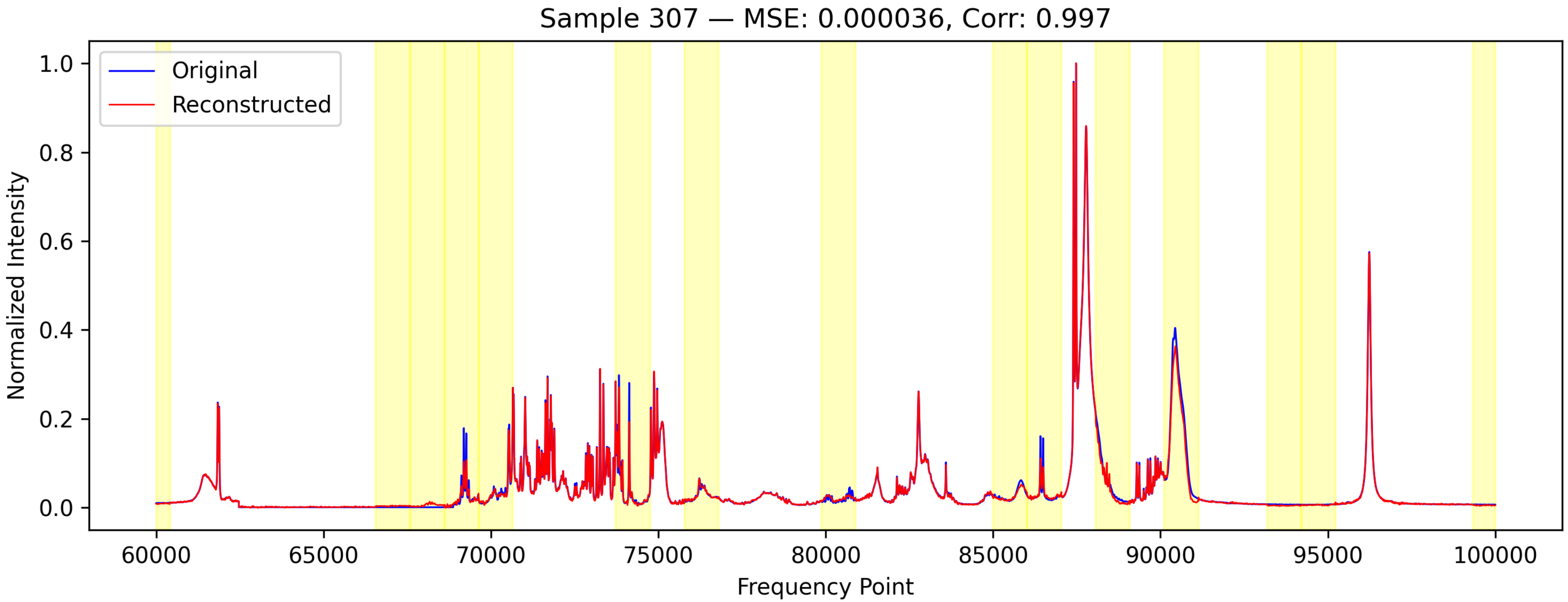
Sample 307 — MSE: 0.000201, Corr: 0.985
Original
Reconstructed
Normalized Intensity
Frequency Point
Sample 486 — MSE: 0.000055, Corr: 0.990
Original
Reconstructed
Normalized Intensity
Frequency Point

15% Masking

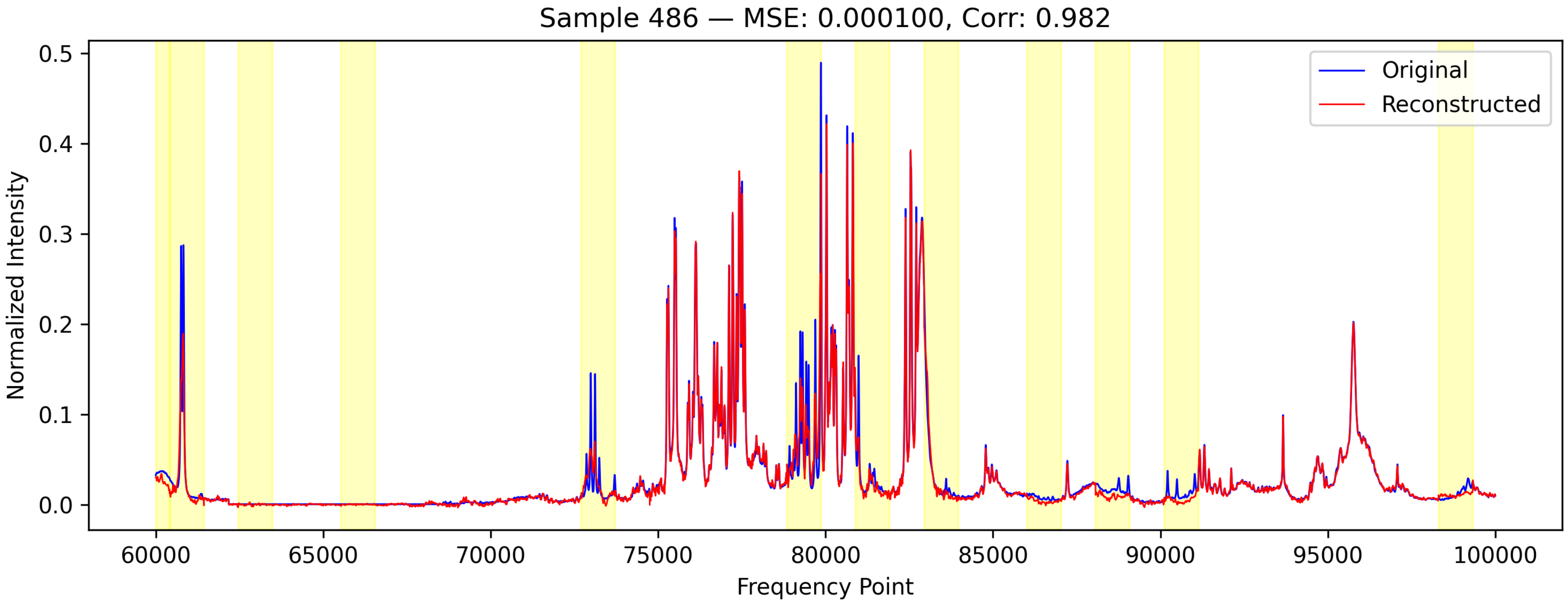


Sample 307 — MSE: 0.000017, Corr: 0.999
Original
Reconstructed
Normalized Intensity
Frequency Point
Sample 486 — MSE: 0.000148, Corr: 0.974
Original
Reconstructed
Normalized Intensity
Frequency Point

25% Masking

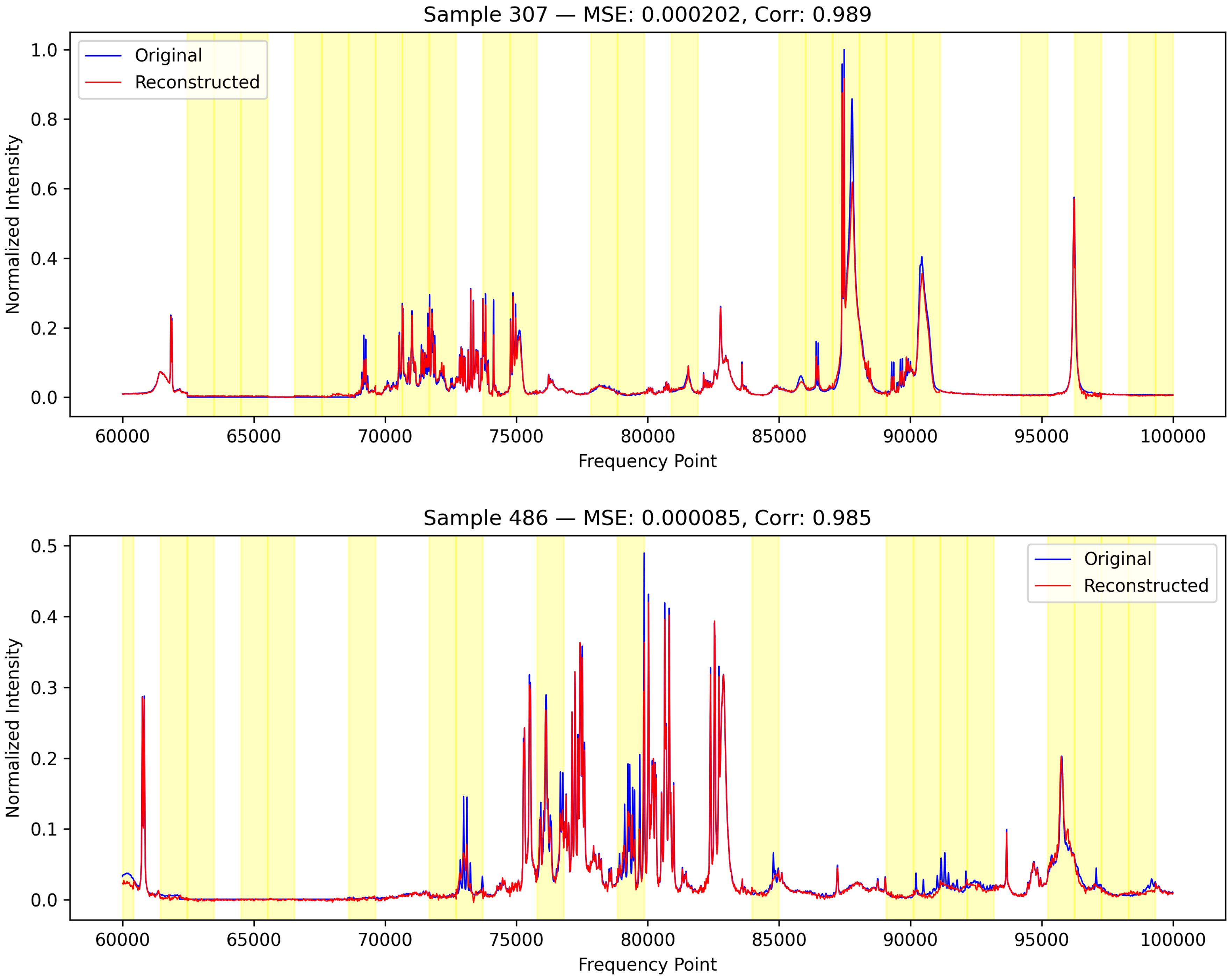


Sample 307 — MSE: 0.000036, Corr: 0.997
Original
Reconstructed
Normalized Intensity
Frequency Point



Sample 486 — MSE: 0.000100, Corr: 0.982
Original
Reconstructed
Normalized Intensity
Frequency Point

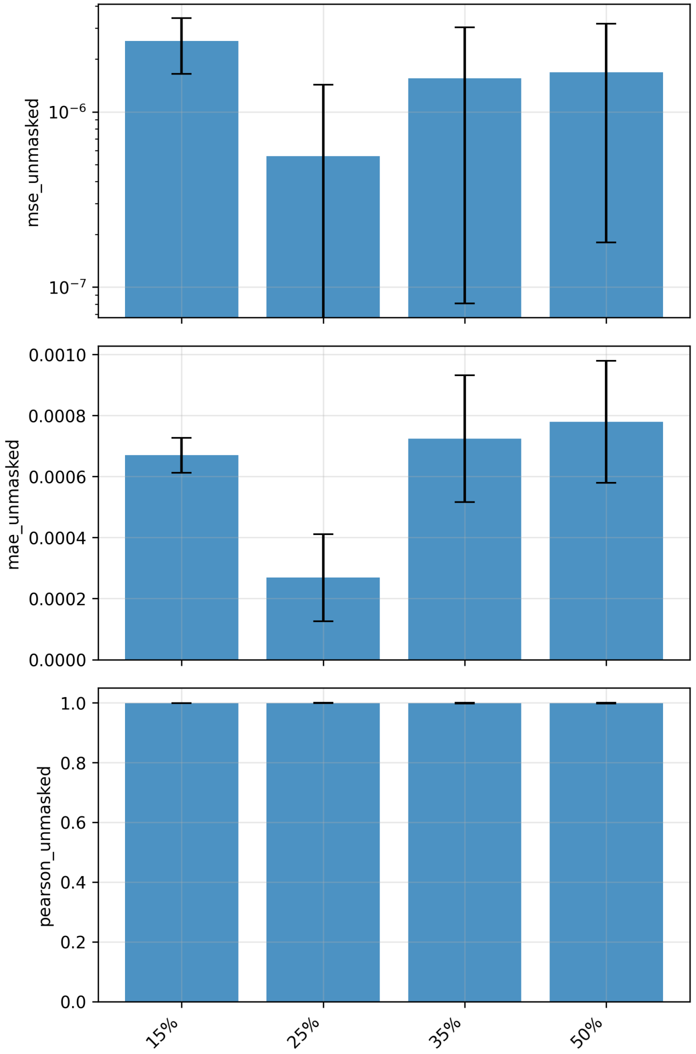
35% Masking



Sample 307 — MSE: 0.000202, Corr: 0.989
Original
Reconstructed
Normalized Intensity
Frequency Point
Sample 486 — MSE: 0.000085, Corr: 0.985
Original
Reconstructed
Normalized Intensity
Frequency Point

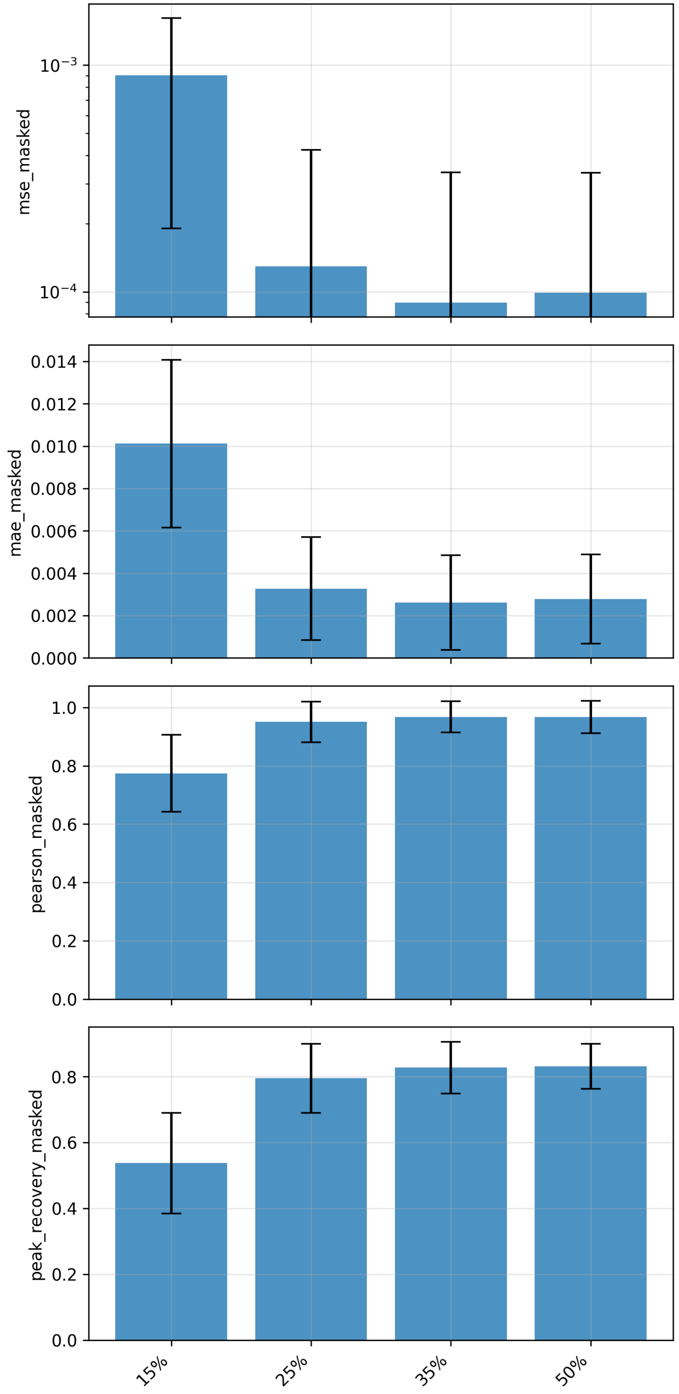
50% Masking

Comparison of Unmasked and Masked regions

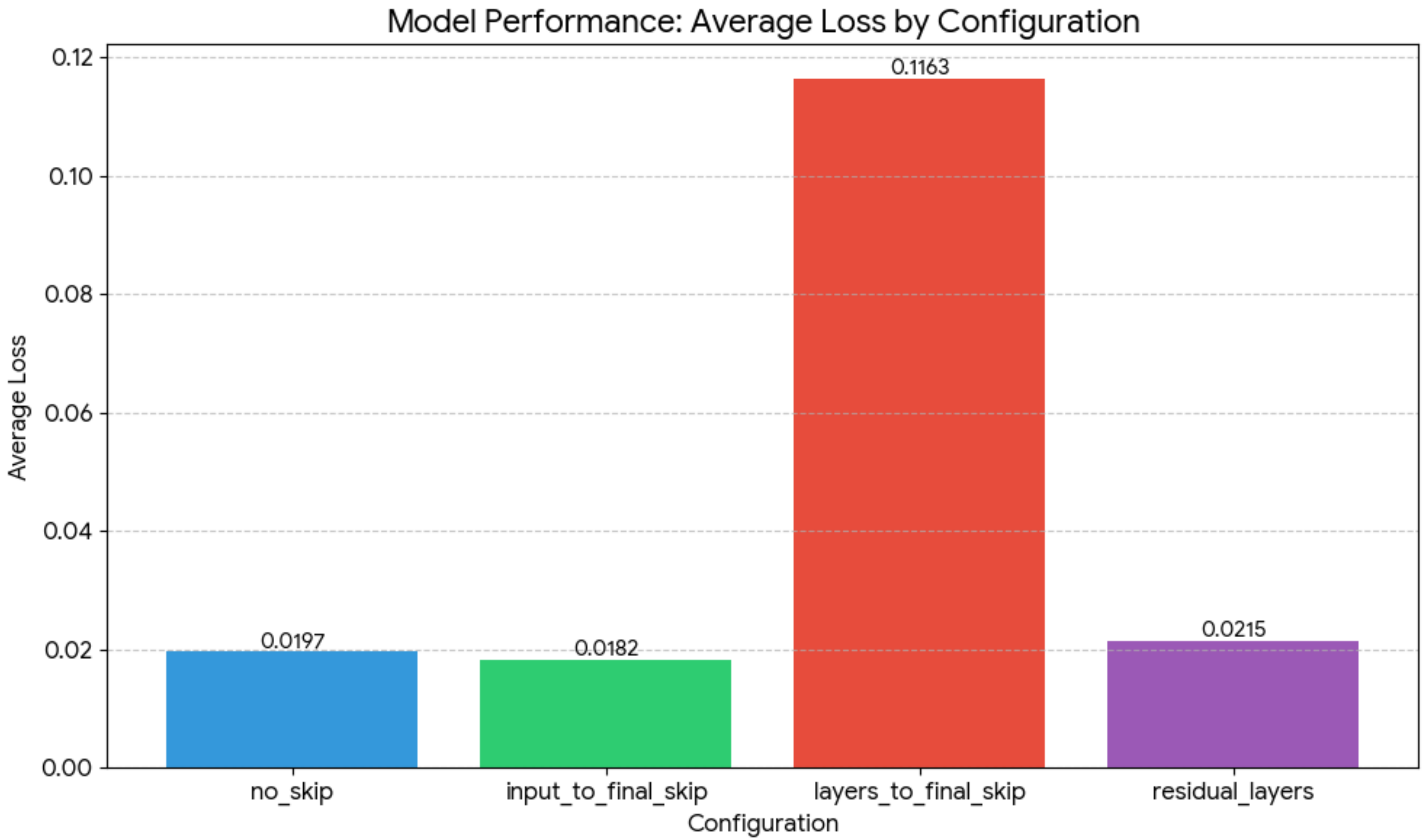


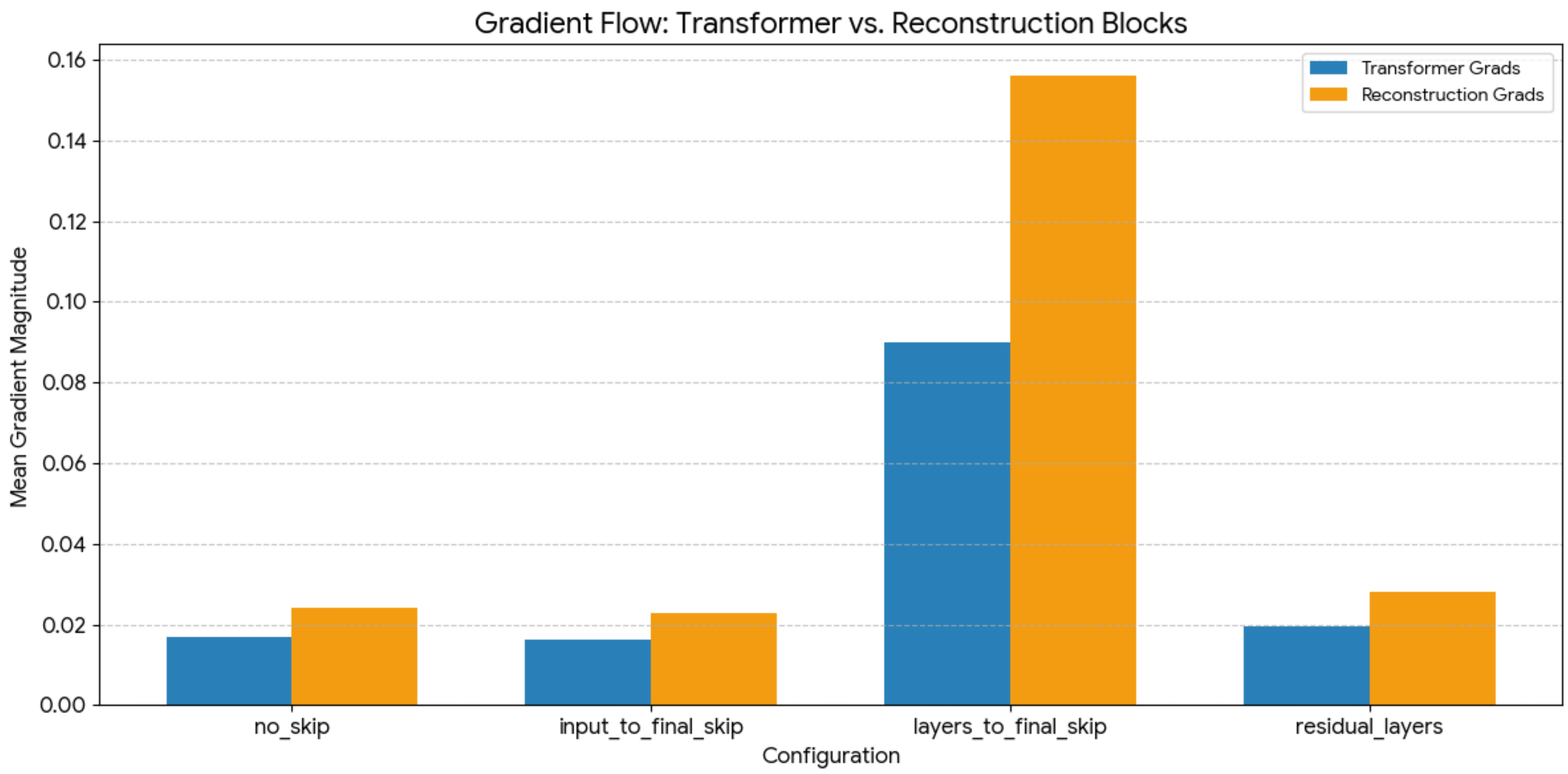
Unmasked

Masked →



Gradient Analysis





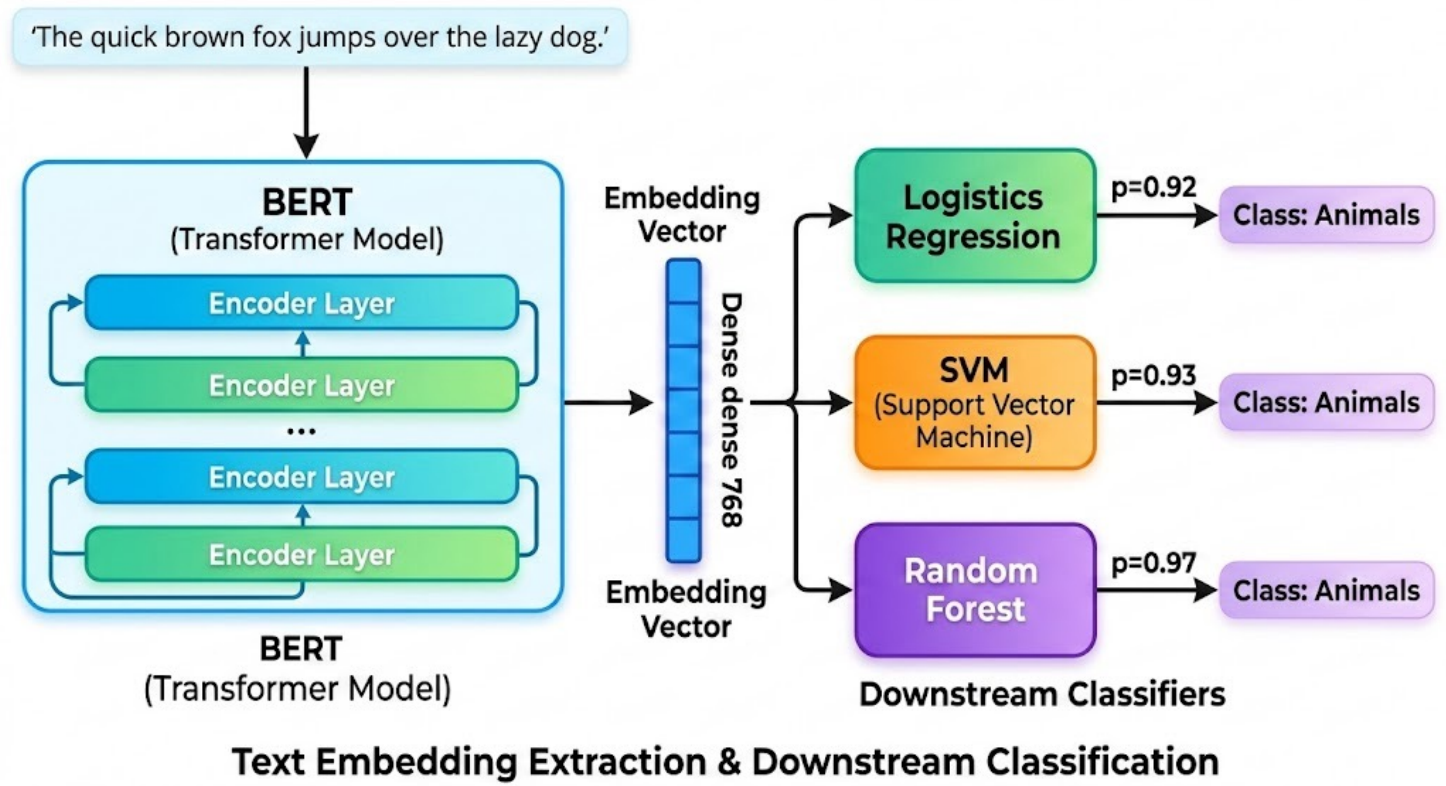
Gradient Flow: Transformer vs. Reconstruction Blocks
Transformer Grads
Reconstruction Grads
Mean Gradient Magnitude
0.00
0.02
0.04
0.06
0.08
0.10
0.12
0.14
0.16
no_skip
input_to_final_skip
layers_to_final_skip
residual_layers
Configuration

Test Data

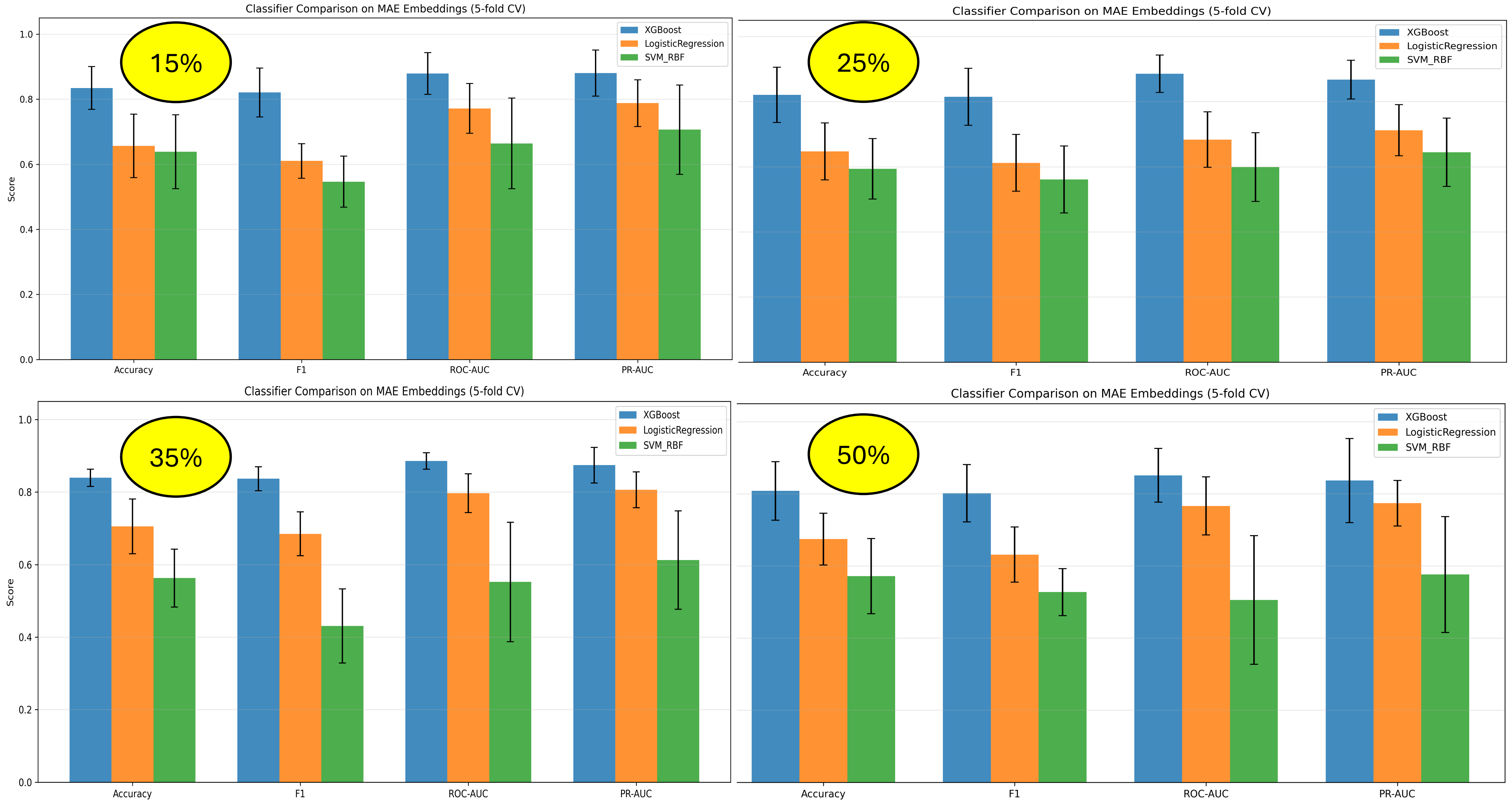
- Traumatic Brain Injury (TBI) samples from Tirupati (thanks to Gayatree)
- CPMG Serum
- ~57 samples
- Collected on 1st Day and 4th Day

Test 1 – Backbone + ML Model

- Three models – SVM, Logistic Regression, XGBoost
- Extract embedded features from our trained model (128)
- Feed into these 3 ML models
- Try to classify – using Cross Validation

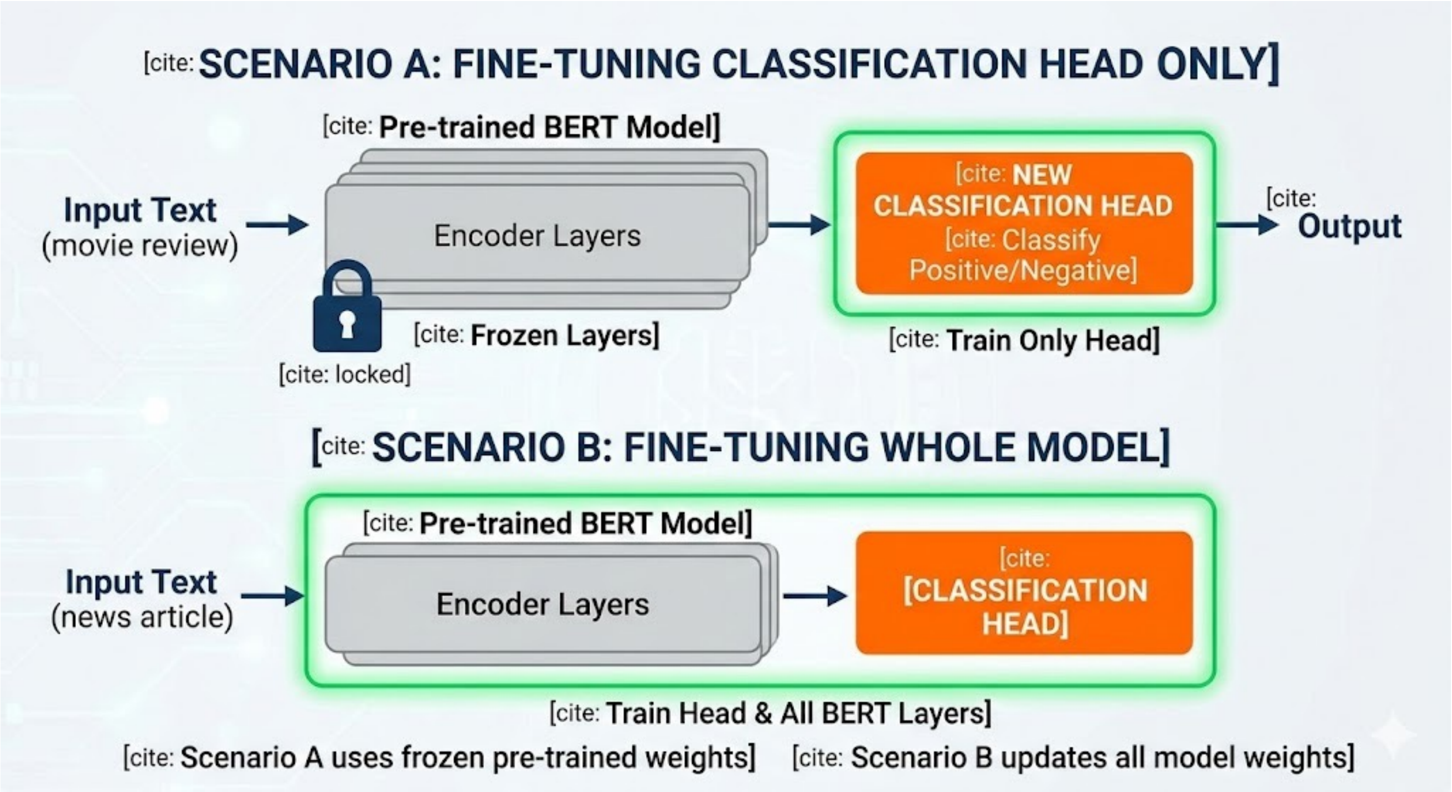


Comparison of 15%, 25%, 35% and 50% Masking – ML Models trained on Embeddings

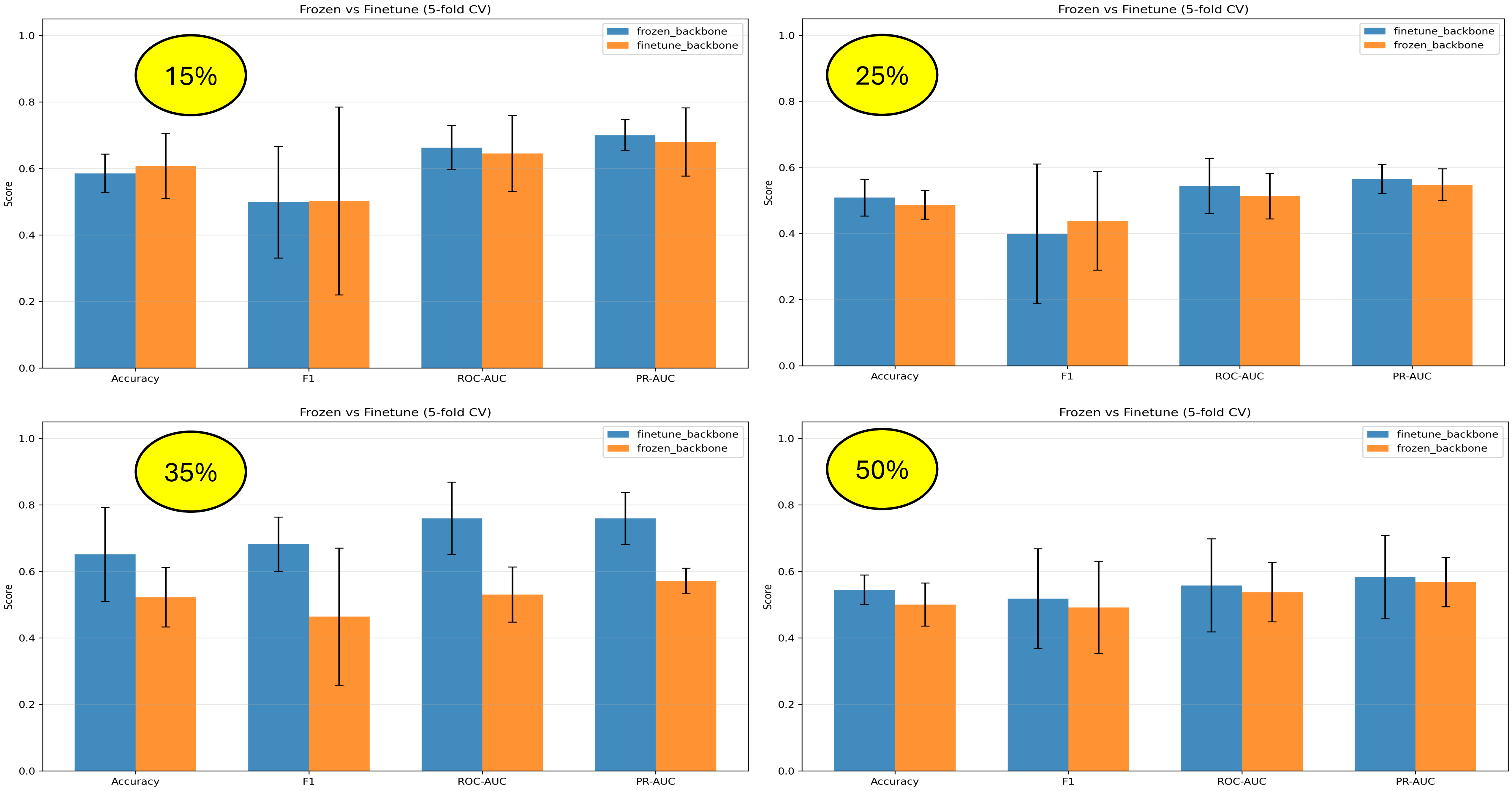


Test 2 – Backbone + Classification head

- Change the final layer of our DNN
- Make it a classification head
- Two methods
  - Freeze the backbone
  - Finetune backbone also
- Try to classify – using Cross Validation



Comparison of 15%, 25%, 35% and 50% Masking – Frozen Backbone vs Finetuning

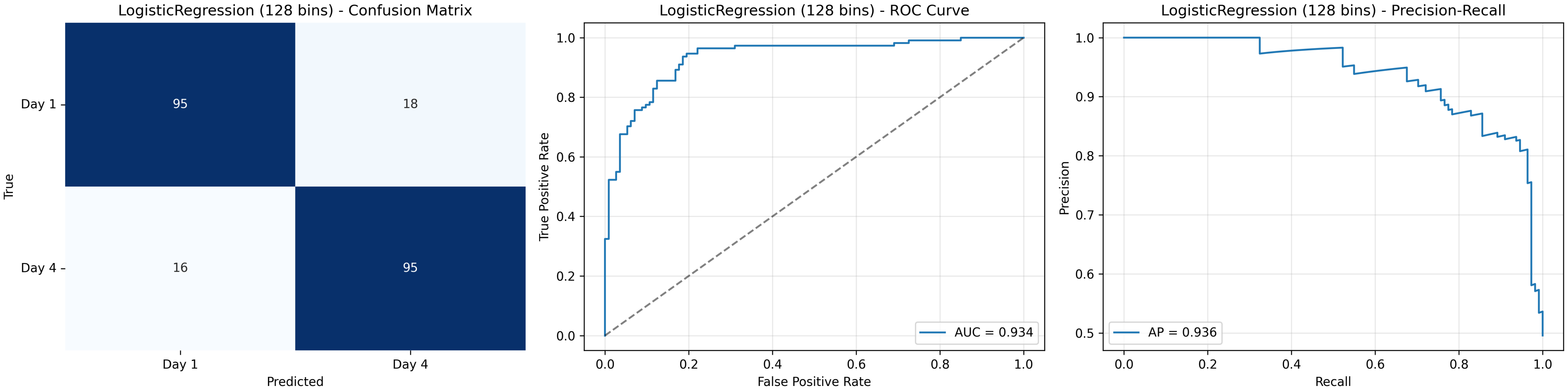


Test 3 – Only ML models

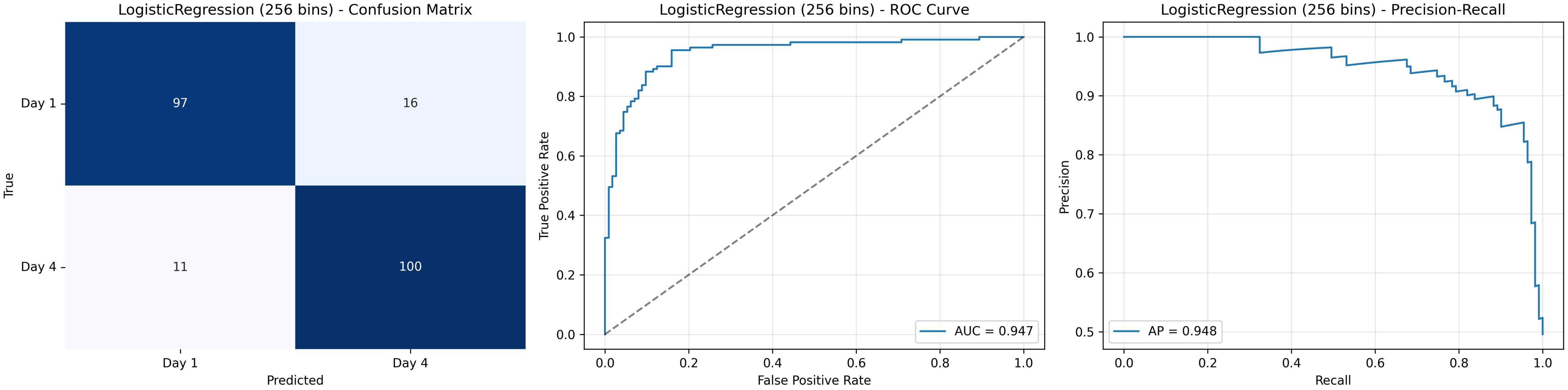
- Bin the spectra into 128 or 256 bins
- Use the 3 ML models directly
- AUC values for each bin to be fed
- Try to classify – using Cross Validation

Logistic Regression – 128 and 256 bins

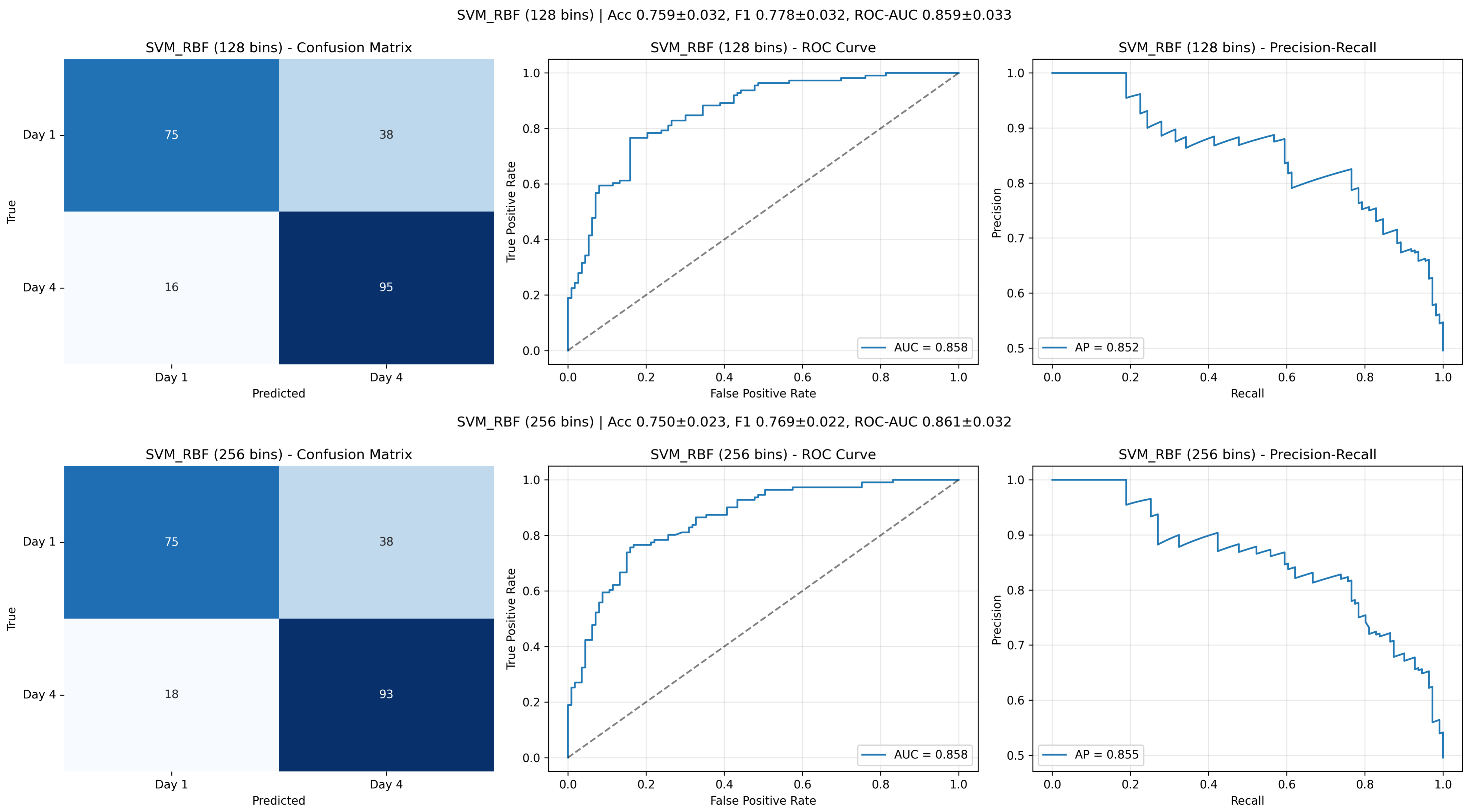
LogisticRegression (128 bins) | Acc 0.848±0.048, F1 0.848±0.052, ROC-AUC 0.935±0.050



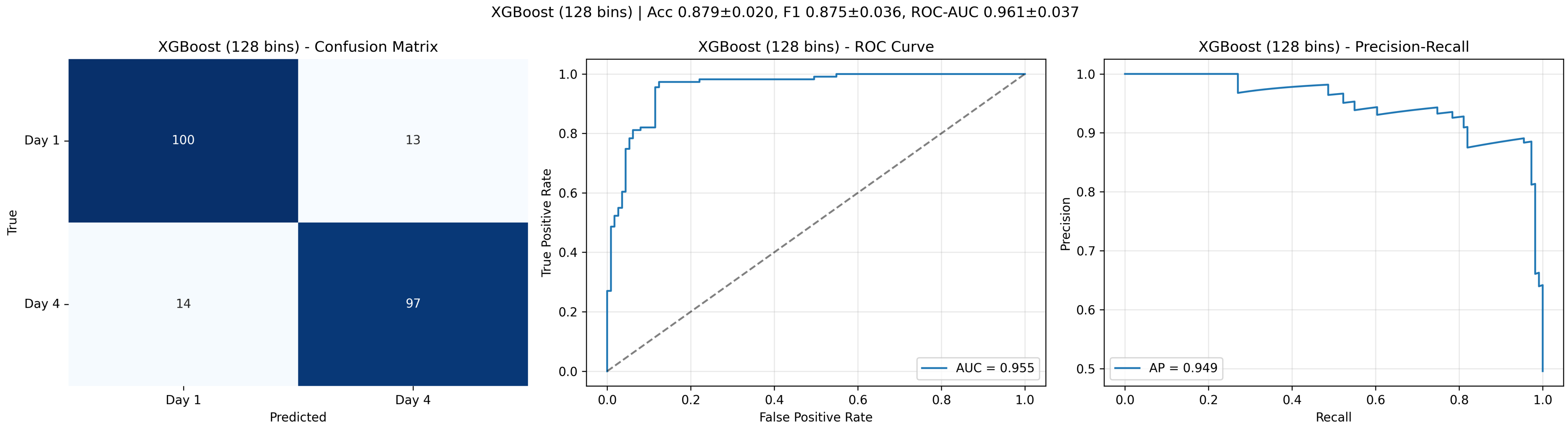
LogisticRegression (256 bins) | Acc 0.880±0.048, F1 0.879±0.057, ROC-AUC 0.947±0.053

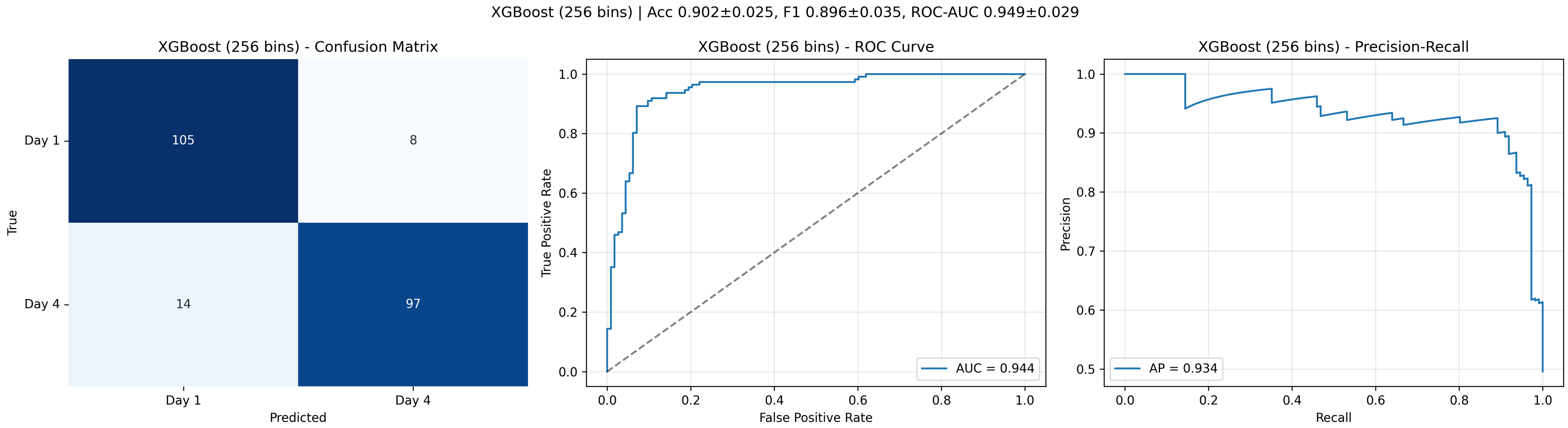


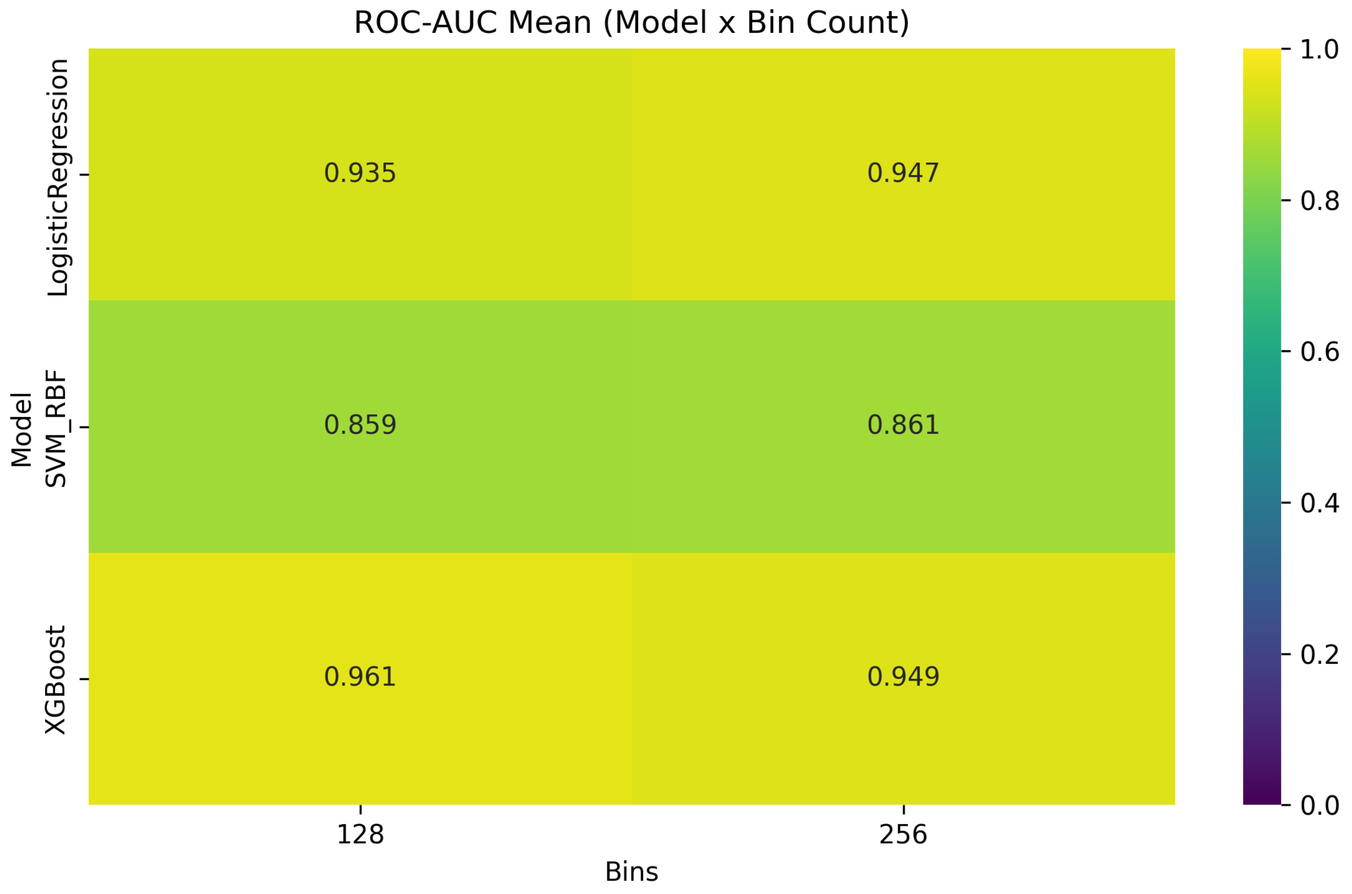
SVM – 128 and 256 bins



XGBoost – 128 and 256 Bins






ROC-AUC Mean (Model x Bin Count)
LogisticRegression
SVM_RBF
XGBoost
Model
0.935
0.947
0.859
0.861
0.961
0.949
128
256
Bins
1.0
0.8
0.6
0.4
0.2
0.0

Thank you all!!